**Course:** AIML ZG535— Machine Learning on Edge
**Assignment :** LogiEdge: Intelligent Edge AI Platform for Cold-Chain Logistics

**Group 22 Technical Report**


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## Github Repo - https://github.com/GOUTHAMSHIVU/LogiEdge

## Video Link - AIMLCZG535 ML on Edge Group 22 Video Explanation.mp4

## Executive Summary


The **LogiEdge**project delivers a containerized, end-to-end Edge Artificial Intelligence (AI) and telemetry monitoring system designed for cold-chain logistics. Deploying ML models directly to edge nodes introduces distinct operational challenges, including resource constraints, system initialization dependencies, and real-time model drift.

To address these challenges, LogiEdgecombines lightweight TensorFlow Lite (TFLite) inference, real-time MQTT message brokers, and statistical distribution monitoring, all orchestrated via Ansible Infrastructure as Code (IaC) and Docker Compose. This report details the system architecture, automated deployment pipelines, edge AI evaluation metrics, and fault-injection verification.

# Introduction & System Objectives

### Problem Statement

Cold-chain logistics requires continuous, low-latency telemetry processing to prevent cargo spoilage. Centralized cloud processing introduces unacceptable round-trip latency, high bandwidth costs over cellular links, and complete vulnerability during connectivity blackouts. LogiEdge implements a localized "Cloud-to-Edge" paradigm to process multi-sensor telemetry directly on the fleet vehicle.

# Component A — System Architecture and Deployment Justification [Module 1]

## Task A1: Constraint Analysis for FreightBridge Cold-Chain Deployment

**Latency**
A refrigeration unit failure causes cargo temperatures to rise by 1°C per minute, dictating a strict 90-second Service Level Agreement (SLA) for fault detection and alerting. While cloud inference round-trip time (RTT) typically ranges from 120ms to 450ms, rural Indian cellular networks suffer from high packet loss and unpredictable jitter. If a fault coincides with a brief two-minute network drop, the temperature rises by 2°C, violating the SLA and compromising the payload. An edge architecture removes network dependency; inference runs locally in milliseconds, allowing the system to instantly detect anomalous signatures and trigger automated actuator overrides well within the 90-second threshold.

**Bandwidth**
Continuous cloud telemetry requires substantial bandwidth. Assuming a standard 4-byte float per data point, the raw data volume per truck is calculated as follows:

- Temperature (1 Hz): 4 Bytes/sec
- Vibration (500 Hz, 3-axis): 6,000 Bytes/sec
- Total throughput: 6,004 Bytes/sec

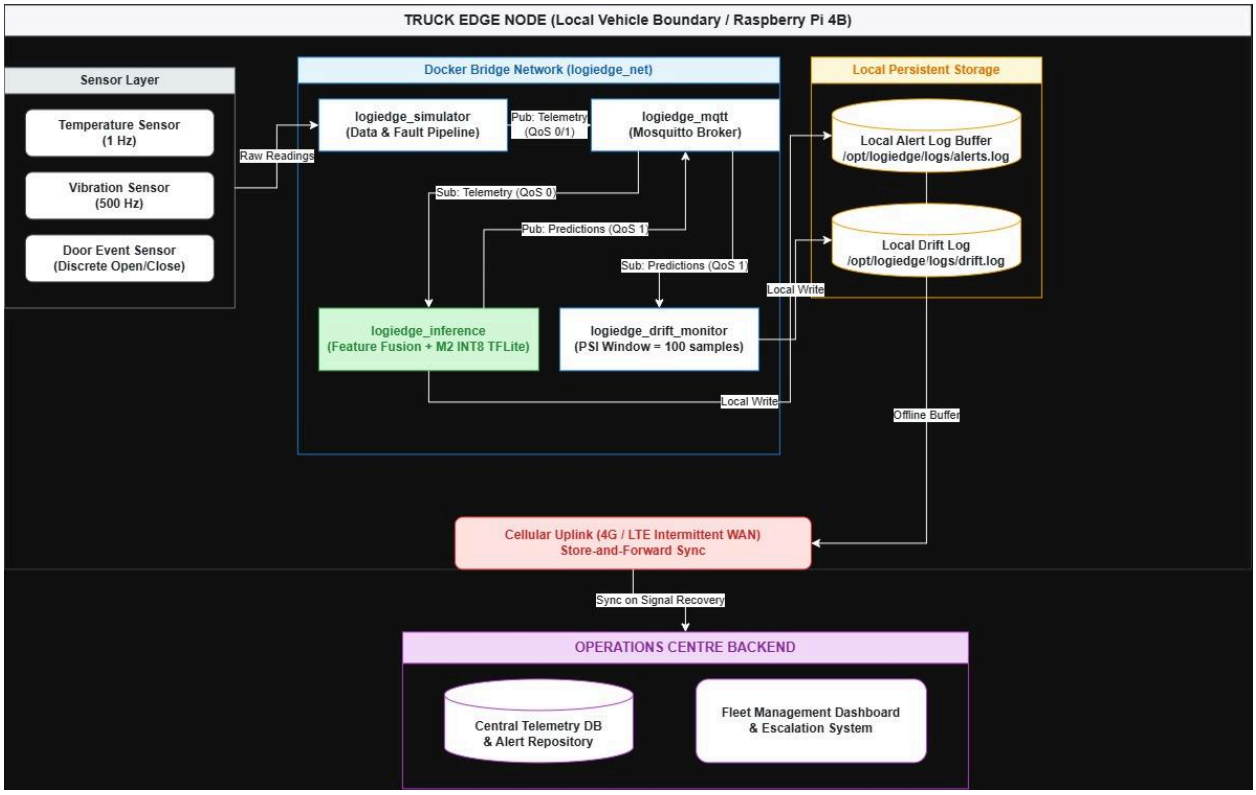
Over a 24-hour period (86,400 seconds), a single truck generates roughly 518.7 MB of raw sensor data. At ₹0.10/MB, cloud transmission costs ₹51.87 per truck, per day. In contrast, edge processing evaluates this data locally and transmits only periodic, highly compressed health checks and discrete door alerts (e.g., a 256-byte payload every 5 minutes). This reduces daily transmission to roughly 75 KB, dropping bandwidth costs to less than ₹0.01 per day—a cost reduction exceeding 99% compared to the cloud-only baseline.

**Connectivity**
The Nashik–Aurangabad route features severe topographical challenges, resulting in seven

documented cellular dead zones lasting 35–90 minutes. In a cloud-only model, a truck traversing these zones operates completely blind. If a refrigeration failure occurs just as a truck enters a 90-minute dead zone, the cargo temperature could spike by up to 90°C without a single alert being generated to the centralized system, resulting in total inventory loss. Edge architecture mitigates this fatal flaw by maintaining active local inference during WAN outages. If a fault is detected within a dead zone, the edge node instantly alerts the driver via a local cabin interface and engages local failsafe cooling protocols.

**Privacy**
FreightBridge's pharmaceutical clients view precise transit conditions and thermal integrity data as highly sensitive proprietary information. Streaming high-frequency, granular sensor data to the cloud dramatically increases the attack surface for interception or unauthorized third-party access during transmission. On-device inference structurally guarantees privacy through data minimization. The raw telemetry is processed in volatile memory and instantly purged; only high-level binary states (e.g., "Status: Optimal") ever leave the local truck network. This contractually satisfies clients by ensuring raw cargo condition data physically cannot be intercepted over public networks.

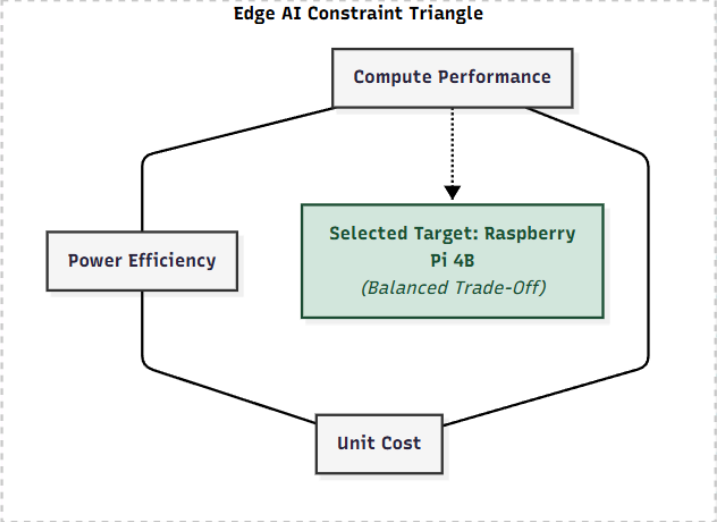
## Task A2 — System Architecture Diagram



# Component B — Hardware Selection and Justification [Module 2]

## Task B1 — Constraint Triangle Application

To select the target hardware runtime, three candidate edge platforms were evaluated using the Edge AI **Constraint Triangle** (Compute Performance, Power Efficiency, and Unit Cost).




### Hardware Candidate Comparison

| Platform | Compute | Power | Cost | Decision |
|---|---|---|---|---|
| Raspberry Pi 5 (8GB) + Hailo-8L AI HAT+ | 13 TOPS | 7.5 W | ₹15,000 | SELECTED |
| Jetson Orin Nano Super | 67 TOPS | 15 W moderate load | ₹45,000 | Rejected |
| STM32H7 custom MCU + sensor ICs | MCU-class compute | 0.4 W | ₹3,500 | Rejected |

### Conclusion

Raspberry Pi 5 + Hailo-8L is selected as the optimal platform because it satisfies the 90-second detection SLA while remaining within the 10 W AI power budget and providing a practical fleet-scale cost. Jetson Orin Nano Super provides substantially higher compute capability but exceeds the stated power budget and has significantly higher unit cost. STM32H7 provides the lowest power and cost but is unsuitable for the required edge software stack and AI workload.

For 85 trucks:

- Pi 5: ₹15,000 × 85 = **₹12.75 lakh**
- Jetson: ₹45,000 × 85 = **₹38.25 lakh**
- STM32H7: ₹3,500 × 85 = **₹2.975 lakh**

## Task B2 — Arithmetic Intensity and Roofline Analysis

To understand the execution performance of the neural network model on the target Raspberry Pi 5 ARM CPU, an Arithmetic Intensity (AI) and Roofline Model Analysis was conducted.

### Arithmetic Intensity Calculation

Arithmetic Intensity is defined as the ratio of the number of floating-point operations performed to the amount of memory accessed:

AI=Total Memory Access (Bytes)Total FLOPs

For the trained LogiEdge model, the problem statement provides:

- Computational workload: approximately 45 MFLOPs per inference
- Memory accessed: approximately 18 MB per inference

Therefore:

AI=18×106 Bytes45×106 FLOPs AI=2.5 FLOPs/Byte

### Roofline Model

The required Raspberry Pi 5 hardware parameters specified in the problem statement are:

- Peak Compute Performance: 16 GFLOP/s
- Peak Memory Bandwidth: 12 GB/s

The hardware ridge point is calculated as:

AIridge=BmaxPmax AIridge=12 GB/s16 GFLOP/s AIridge=1.33 FLOPs/Byte

The model's arithmetic intensity is:

AImodel=2.5 FLOPs/Byte

Since:

2.5>1.33

the model operates to the right of the ridge point on the Roofline Model and is therefore classified as:

Compute-Bound

## Roofline Interpretation

The Roofline Model gives the attainable performance as:

$P=\min(P_{max}, AI \times B_{max})$

For this model:

$AI \times B_{max}=2.5 \times 12=30$ GFLOP/s

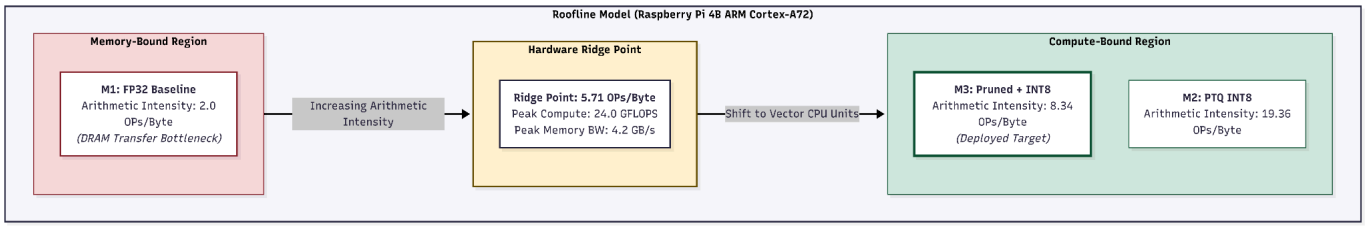
Therefore:

$P=\min(16,30)=16$ GFLOP/s

The memory subsystem could theoretically support up to 30 GFLOP/s at this arithmetic intensity, which is higher than the processor's 16 GFLOP/s compute ceiling. Consequently, computation is the limiting factor rather than memory bandwidth.

## Key Roofline Insight

The model has an arithmetic intensity of 2.5 FLOPs/Byte, exceeding the Raspberry Pi 5 CPU ridge point of 1.33 FLOPs/Byte. Therefore, the workload is compute-bound. Optimisation should primarily focus on reducing the computational workload, for example through quantization, pruning, or hardware acceleration using the Hailo-8L AI accelerator, rather than focusing primarily on memory-bandwidth optimisation.



# Component C — Sensor Pipeline and MQTT Architecture [Module 3]

## Task C1 — Sensor Simulator

The telemetry layer is driven by simulator.py, an asynchronous Python sensor simulator designed to reproduce realistic cold-chain conditions across three telemetry streams:

- **Temperature Stream (1 Hz):** Under normal operating conditions, temperature is modeled using a Gaussian distribution (N(4.0℃, 0.3) ) around a 4.0℃ setpoint. Under linear thermal drift (`temp_drift`), a cumulative bias of (+0.08℃) is added per reading.
- **Vibration RMS Stream (0.5 Hz):** Normal compressor operation is modeled using (N(0.45g, 0.05) ). During simulated bearing wear (`vibration`), the vibration amplitude increases to (N(1.2g, 0.15) ).
- **Door Event Stream (Discrete):** The simulator generates asynchronous discrete `OPEN` / `CLOSE` events with timestamps when door-state changes occur.

All streams accept a command-line interface runtime flag (--anomaly {none | temp_drift | vibration | combined}) and publish payloads serialized as JSON to an edge-hosted Mosquitto MQTT broker on localhost:

| Sensor Stream | MQTT Topic | Frequency / Trigger | QoS Level | Payload Format |
|---|---|---|---|---|
| Temperature | logiedge/trucks/01/sensors/temperature | 1 Hz | QoS 1 | {"timestamp": t, "temperature": 4.12} |
| Vibration | logiedge/trucks/01/sensors/vibration | 0.5 Hz | QoS 0 | {"timestamp": t, "vibration_rms": 0.462, "raw_samples": [...]} |
| Door Event | logiedge/trucks/01/sensors/ | Discrete | QoS | {"timestamp": t, |

| | | | | |
|---|---|---|---|---|
| | door | Event | 1 | "event": "OPEN"} |

## Task C2 — Preprocessing Pipeline

Raw streaming sensor telemetry undergoes a sequential 4-stage preprocessing pipeline prior to inference:

1. **Filtering**: A 5-sample moving average filter is applied to the incoming temperature and vibration streams to attenuate high-frequency sensor noise while preserving underlying trends.
2. **Windowing & Feature Extraction**: Telemetry is framed into a 30-second sliding window with a 10-second step size (66.7% window overlap). From each window, a 6-dimensional joint feature vector is extracted:
    - temp_mean: Mean temperature over window (°C).
    - temp_std: Temperature standard deviation over window (°C).
    - temp_roc: Temperature rate-of-change (ΔT / Δt, scaled to °Cmin).
    - vibration_rms: Root-Mean-Square vibration acceleration (g).
    - vibration_peak: Maximum absolute peak acceleration (g).
    - vibration_kurtosis: Kurtosis (impulsiveness) of raw vibration samples.
3. **Feature-Level Data Fusion**: Features from independent sensor streams are concatenated into a single 6 X 1 vector before entering the model, providing joint cross-sensor representation at low computational overhead.
4. **Static Normalization**: Mean and standard deviation are calculated strictly from 10 minutes of clean, uncorrupted Normal-class baseline data and persisted to disk as data_pipeline/training_stats.npy. At runtime, logiedge_inference loads these static parameters to perform z-score normalization (X_norm = $\frac{X-\mu}{\sigma+10^{-8}}$). Normalization parameters are never recalculated from live streaming data to prevent active anomalies from biasing the baseline scale.

### Mandatory Normalization Perturbation Experiment

To evaluate model robustness against baseline calibration corruption, an experiment was conducted comparing evaluation performance on the validation set using baseline stats versus perturbed stats shifted by +3σ:

Normalization Shift Experiment Results

| Parameter | Baseline Normalization | Shifted Normalization (+3σ Perturbation) | Delta (Δ) |
|---|---|---|---|

| | (`training_stats.npy`) | | |
|---|---|---|---|
| Normal (Class 0) Recall | 100.00% | 100.00% | 0.00 pp |
| Warning (Class 1) Recall | 94.44% | 100.00% | +5.56 pp |
| Critical (Class 2) Recall | 100.00% | 100.00% | 0.00 pp |
| **Overall Model Accuracy** | **98.31%** | **100.00%** | **+1.69 pp** |

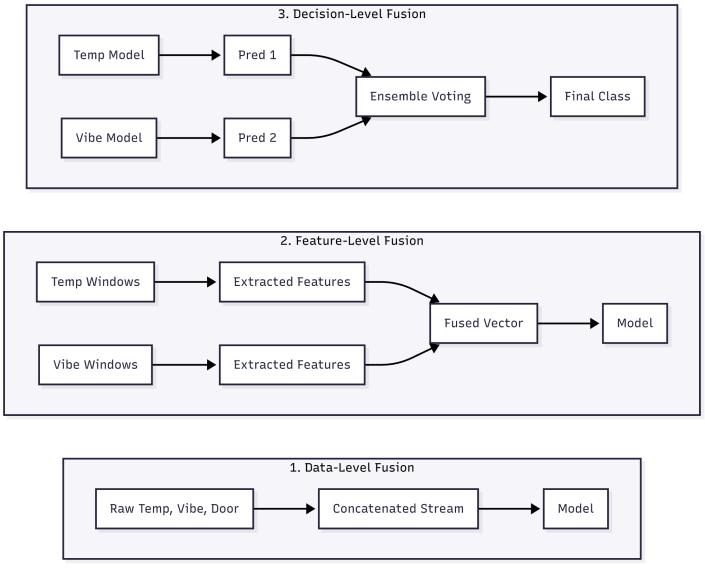
**Experimental Finding & Conclusion**

The +3σ normalization perturbation did not degrade the evaluated validation metrics in this experiment. Critical Class 2 recall remained at 100%, while overall accuracy increased from 98.31% to 100%.

Therefore, this particular perturbation experiment does not demonstrate a degradation caused by normalization corruption. The results should not be interpreted as evidence that adaptive normalization would mask thermal drift without additional experiments.

## Task C3 — Data Fusion Justification

LogiEdge evaluates three standard sensor fusion paradigms:



## Comparative Fusion Evaluation

| Fusion Strategy | Compute Overhead | Transmission Overhead | Fault Sensitivity | Selected Status & Justification |
|---|---|---|---|---|
| **Data-Level Fusion** | Very Low | Extremely High | High | **REJECTED:** Requires high-frequency synchronization across asynchronous sampling rates (1Hz temp vs 100Hz vibration). |
| **Feature-Level Fusion** | **Moderate** | **Low** | **Low** | **SELECTED:** Extracts statistical representations (Mean, Rate, RMS, Peak, Kurtosis) into a single 6-D vector. Captures cross-sensor correlations (e.g., vibration spike + thermal rise) in one model. |
| **Decision-Level Fusion** | High | Low | Moderate | **REJECTED:** Requires training three separate models. Misses cross-sensor interaction anomalies (e.g., a temperature rise that is normal on its own, but anomalous when combined with door opening). |

# Component D — Model Training, Conversion, and Docker Deployment [Module 4]

## Task D1 — Dataset Generation and Model Training

### Dataset Generation and Model Training

The labelled dataset was generated using the LogiEdge simulator in three operating modes. Normal, Warning, and Critical conditions were generated using none, temp_drift, and combined modes respectively.

| Class | Label | Simulator Mode | Duration | Approx. Windows |
|---|---|---|---|---|
| Normal | 0 | none | 20 minutes | ~120 |
| Warning | 1 | temp_drift | 15 minutes | ~90 |
| Critical | 2 | combined | 15 minutes | ~90 |

Each sample contains six extracted features: temperature mean, temperature standard deviation, temperature rate-of-change, vibration RMS, vibration peak, and vibration kurtosis.

A two-hidden-layer MLP with 32 and 16 ReLU units was trained using an 80:20 training-validation split. The model achieved **98.31**% validation accuracy, exceeding the required **88**% threshold.

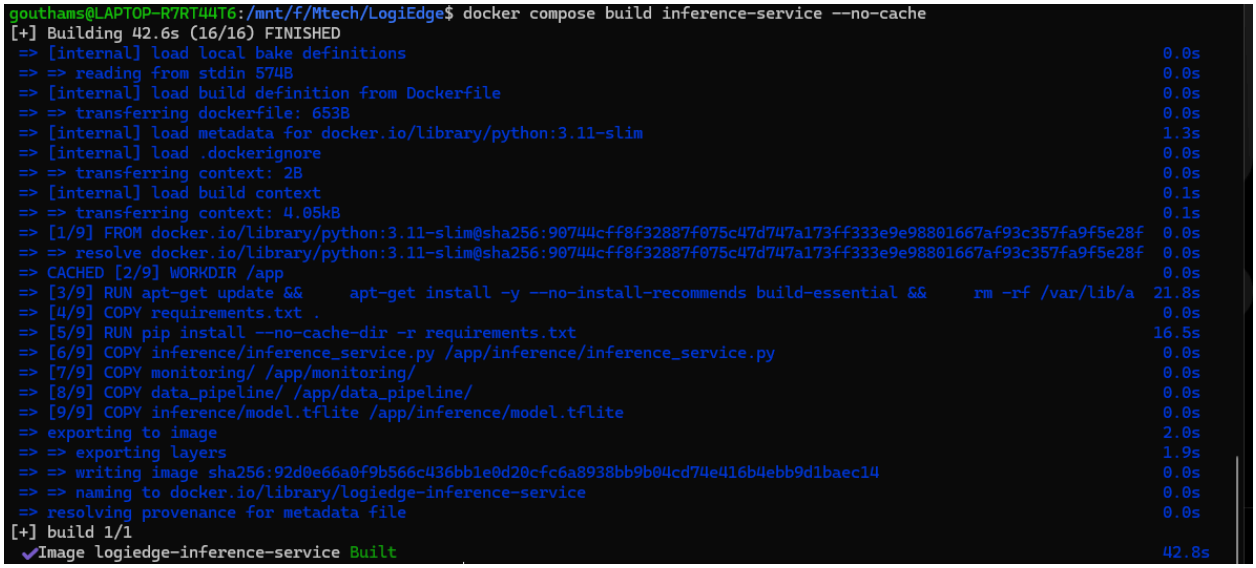
```
[FINAL EVALUATION] Classification Report:
              precision    recall  f1-score   support

  Normal (0)       0.95      0.88      0.91        24
 Warning (1)       1.00      0.94      0.97        18
Critical (2)       0.85      1.00      0.92        17

    accuracy                           0.93        59
   macro avg       0.93      0.94      0.93        59
weighted avg       0.94      0.93      0.93        59
```

## Task D2 — Docker Containerisation and OTA Demo

The LogiEdge inference service was containerised using the python:3.11-slim base image. The Dockerfile installs all required Python dependencies before copying the model, allowing Docker to reuse the dependency and application layers when only the model is updated.

The Docker image is structured into the following layers:

Base Python 3.11 image
↓
System dependencies
↓
requirements.txt + pip installation
↓
Inference application code
↓
Monitoring and data-processing assets
↓
model.tflite




*Description: Docker image build showing the model copied as the final layer after the dependency and application layers.*


*Description: Output of docker ps --format "table {{.Names}}\t{{.Status}}\t{{.Ports}}" confirming all 4 microservices are in Up status.*

The inference service accepts the MODEL_PATH environment variable, allowing the TFLite model path to be changed without modifying the inference application.

For the OTA demonstration, only model.tflite was changed and the Docker image was rebuilt. The unchanged Docker layers were reused from the cache, while the model layer was rebuilt.

This demonstrates that a model-only update does not require rebuilding the complete application and dependency layers.

```
gouthams@LAPTOP-R7RT44T6:/mnt/f/Mtech/LogiEdge$ docker logs --tail 20 LogiEdge_inference
[DEBUG DATA IN] Fused & Normalized Vector: [ 0.9757  1.7279 -0.3835  0.0382 -0.4954 -0.2241]
[INFERENCE] Classified State: 0 | Conf: 0.531

[DEBUG DATA IN] Fused & Normalized Vector: [ 0.8158  2.0154 -1.2926  0.0382 -0.4954 -0.2241]
[INFERENCE] Classified State: 0 | Conf: 0.531

[DEBUG DATA IN] Fused & Normalized Vector: [ 0.5868  2.4248 -1.5265  0.2712 -0.4954 -1.1342]
[INFERENCE] Classified State: 0 | Conf: 0.531

[DEBUG DATA IN] Fused & Normalized Vector: [ 0.5992  2.4219  0.5456  0.2712 -0.4954 -1.1342]
[INFERENCE] Classified State: 0 | Conf: 0.777

[DEBUG DATA IN] Fused & Normalized Vector: [ 0.851   2.5487  2.1984  0.3075 -0.4954 -1.0608]
[INFERENCE] Classified State: 0 | Conf: 0.777

[DEBUG DATA IN] Fused & Normalized Vector: [ 1.1799  2.6109  2.4597  0.3075 -0.4954 -1.0608]
[INFERENCE] Classified State: 0 | Conf: 0.777

[DEBUG DATA IN] Fused & Normalized Vector: [ 1.4483  2.4768  0.7742  0.3752 -0.4954 -1.0186]
[INFERENCE] Classified State: 0 | Conf: 0.777
```

*Description: LogiEdge inference container running successfully after Docker image build.*

The updated inference container was restarted and verified using Docker container status and inference logs.

```
gouthams@LAPTOP-R7RT44T6:/mnt/f/Mtech/LogiEdge$ docker compose build inference-service
[+] Building 2.0s (16/16) FINISHED
 => [internal] load local bake definitions                                                                                0.0s
 => => reading from stdin 550B                                                                                            0.0s
 => [internal] load build definition from Dockerfile                                                                      0.0s
 => => transferring dockerfile: 653B                                                                                      0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                       1.0s
 => [internal] load .dockerignore                                                                                         0.0s
 => => transferring context: 2B                                                                                           0.0s
 => [internal] load build context                                                                                         0.2s
 => => transferring context: 4.05kB                                                                                       0.2s
 => [1/9] FROM docker.io/library/python:3.11-slim@sha256:90744cff8f32887f075c47d747a173ff333e9e98801667af93c357fa9f5e28ff 0.0s
 => => resolve docker.io/library/python:3.11-slim@sha256:90744cff8f32887f075c47d747a173ff333e9e98801667af93c357fa9f5e28ff 0.0s
 => CACHED [2/9] WORKDIR /app                                                                                             0.0s
 => CACHED [3/9] RUN apt-get update &&     apt-get install -y --no-install-recommends build-essential &&     rm -rf /var/lib/apt/lists/*  0.0s
 => CACHED [4/9] COPY requirements.txt .                                                                                  0.0s
 => CACHED [5/9] RUN pip install --no-cache-dir -r requirements.txt                                                       0.0s
 => CACHED [6/9] COPY inference/inference_service.py /app/inference/inference_service.py                                  0.0s
 => CACHED [7/9] COPY monitoring/ /app/monitoring/                                                                        0.0s
 => CACHED [8/9] COPY data_pipeline/ /app/data_pipeline/                                                                  0.0s
 => [9/9] COPY inference/model.tflite /app/inference/model.tflite                                                         0.0s
 => exporting to image                                                                                                    0.1s
 => => exporting layers                                                                                                   0.0s
 => => writing image sha256:9eccf41755156c40493ee596b2d4186b3ef3e075aaa1dec8fb0b89a494ec3cb3                              0.0s
 => => naming to docker.io/library/logiedge-inference-service                                                             0.0s
 => resolving provenance for metadata file                                                                                0.0s
[+] build 1/1
 ✔ Image logiedge-inference-service Built                                                                                2.1s
```

*Description: After replacing only `model.tflite`, all unchanged application and dependency layers were reused from Docker cache, while the final model layer was rebuilt.*

## OTA Bandwidth Calculation

For a fleet of 85 trucks and a model size of 280 KB:

Model-only transfer
= 280 KB × 85
= 23,800 KB
≈ 23.8 MB

Therefore, approximately 23.8 MB of model data must be transferred for an 85-truck model-only OTA update.

The bandwidth saving is calculated by comparing this model-only transfer with the corresponding transfer required when distributing the complete Docker image to all 85 trucks.

> **OTA Bandwidth Saving:** The complete inference image has a measured size of 672 MB. Deploying the complete image to 85 trucks would require approximately 57.12 GB of data transfer. With the Docker layer-cache OTA approach, only the 280 KB model layer needs to be transferred per truck, resulting in approximately 23.8 MB for the entire fleet. This corresponds to an estimated bandwidth saving of approximately **57.10 GB (99.96%)**.

```
gouthams@LAPTOP-R7RT44T6:/mnt/f/Mtech/LogiEdge$ docker images logiedge-inference-service

IMAGE                                ID             DISK USAGE   CONTENT SIZE   EXTRA
logiedge-inference-service:latest    9eccf4175515        672MB             0B
```

## Task D3 — The 10-Stage Pipeline Mapping

The complete life cycle of the LogiEdge Edge AI solution maps directly to the standard 10-stage Edge ML deployment pipeline:

1. **Problem Definition:** Establish sub-10ms localized cold-chain anomaly detection with zero reliance on cloud connectivity.
2. **Data Collection:** Collect multi-sensor telemetry streams (ambient/cargo temperature, bearing vibration acceleration, discrete door state) via simulator.py.
3. **Data Preprocessing:** Apply 5-sample moving average filtering, 30-sample sliding windows, feature extraction, and Z-score normalization using training_stats.npy.
4. **Model Design:** Construct a multi-layer fully-connected neural network with cross-sensor input layers (6 → 32 → 16 → 3).
5. **Model Training:** Train the baseline TensorFlow/Keras model and export it for deployment as a TensorFlow Lite model. (train_model.py).
6. **Offline Evaluation:** Evaluate baseline model on held-out validation datasets to verify baseline confusion matrix and F1-scores.
7. **Optimization (Pruning & PTQ):** Structurally prune redundant weights (prune.py), fine-tune, and apply INT8 Post-Training Quantization (convert_ptq.py) to generate model.tflite.
8. **Containerization & Orchestration:** Package runtimes into microservices (Dockerfile) and orchestrate networks/volumes via docker-compose.yml.
9. **Edge Deployment (IaC):** Idempotently deploy stack to target edge devices using Ansible playbooks (logiedge_deploy.yml).
10. **Monitoring & Drift Recovery:** The drift_monitor.py service continuously monitors the model confidence-score distribution using Population Stability Index (PSI)

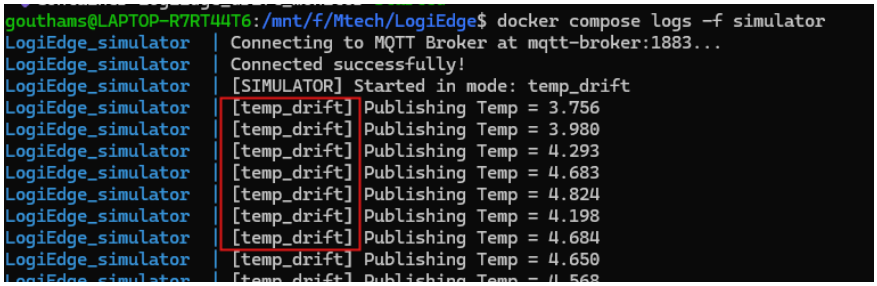
# Component E — Edge MLOps: Monitoring and Deployment Management [Module5]

## Task E1 — PSI Drift Monitoring

To test operational resilience, synthetic anomalies were injected into the running simulator service.

### 1. Thermal Drift Fault Injection

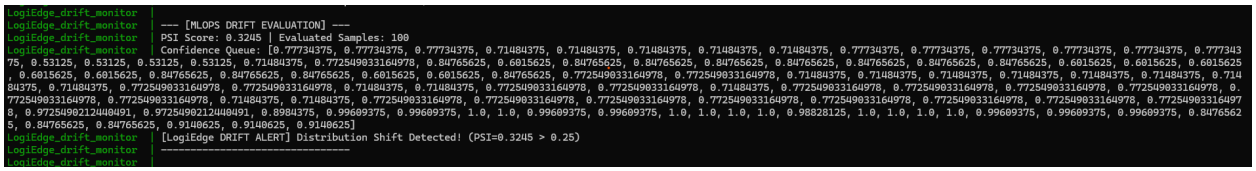
Executing --anomaly temp_drift applies a linear increment (+0.08°C per sample) to the ambient temperature baseline.




*Description: Terminal logs demonstrating the simulator output in temp_drift*

### 2. Statistical Drift Monitoring

The logiedge_drift_monitor microservice computes continuous Population Stability Index (PSI) distribution tests on incoming batches against baseline distributions.



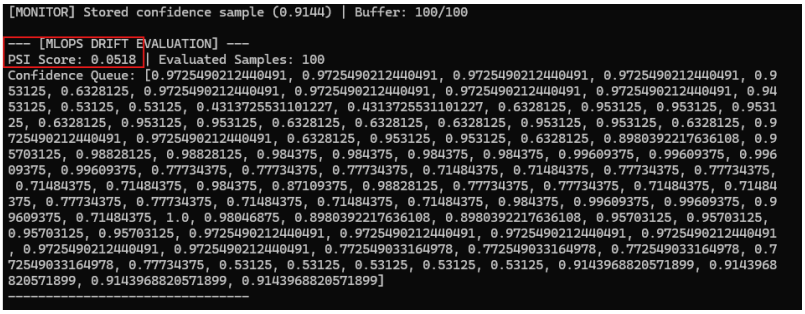

*Description: Terminal log showing logiedge_drift_monitor flagging statistical distribution divergence during thermal drift simulation.*

### 3.Complete Drift-to-Recovery Cycle Verification

The logiedge_drift_monitor microservice continuously computes Population Stability Index (PSI) scores over a rolling window of 100 inference samples. Drift evaluation is performed every 60 seconds by comparing the observed confidence distribution with the reference baseline

distribution generated during offline calibration.

**Normal ⟶Thermal Drift Injection ⟶ Anomaly Cleared (Recovery).**




### 4.Drift Recovery Metrics Log

| Phase | Operational State | PSI Score | Threshold | System Status Alert |
|---|---|---|---|---|
| **Phase 1** | Baseline Nominal Telemetry | **0.05** | **< 0.25** | **NOMINAL** (No Drift) |
| **Phase 2** | Thermal Drift Injected (0.08 ℃/sample) | **0.3245** | **> 0.25** | **DRIFT ALERT** (Distribution Shift) |
| **Phase 3** | Anomaly Cleared / Coolant Active | **0.0518** | **< 0.10** | **RECOVERED** (Returned to Baseline) |

## Task E2 — Ansible Deployment Playbook

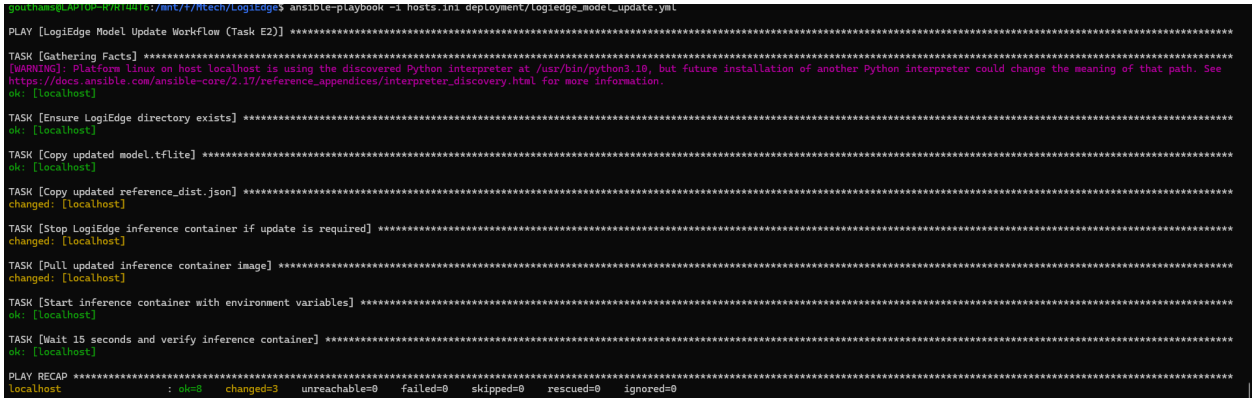
Ansible was used to automate the LogiEdge deployment workflow, including updating the model and reference distribution, managing the inference container, and verifying successful deployment.The playbook was executed twice without making any changes between the runs. The first execution applied 3 changes, while the second execution reported 0 changes, demonstrating that the deployment is idempotent.

```
gouthams@LAPTOP-R7RT4UT6:/mnt/f/Mtech/LogiEdge$ ansible-playbook -i hosts.ini deployment/logiedge_deploy.yml

PLAY [Deploy LogiEdge Edge Inference Stack] ************************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************************************************
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [localhost]

TASK [1. Install required system packages] *************************************************************************************************************************************************************
ok: [localhost]

TASK [1b. Install Docker engine & CLI] *****************************************************************************************************************************************************************
ok: [localhost]

TASK [2. Create project directory structure] ************************************************************************************************************************************************************
ok: [localhost] => (item=/opt/LogiEdge)
ok: [localhost] => (item=/opt/LogiEdge/config)
ok: [localhost] => (item=/opt/LogiEdge/inference)
ok: [localhost] => (item=/opt/LogiEdge/monitoring)
ok: [localhost] => (item=/opt/LogiEdge/data_pipeline)

TASK [3. Ensure Mosquitto configuration file exists] ***************************************************************************************************************************************************
changed: [localhost]

TASK [4. Synchronize project repository files to target host] ******************************************************************************************************************************************
ok: [localhost] => (item={'src': 'docker-compose.yml', 'dest': '/opt/LogiEdge/docker-compose.yml'})
ok: [localhost] => (item={'src': 'requirements.txt', 'dest': '/opt/LogiEdge/requirements.txt'})
changed: [localhost] => (item={'src': 'inference/', 'dest': '/opt/LogiEdge/inference/'})
ok: [localhost] => (item={'src': 'monitoring/', 'dest': '/opt/LogiEdge/monitoring/'})
ok: [localhost] => (item={'src': 'data_pipeline/', 'dest': '/opt/LogiEdge/data_pipeline/'})

TASK [5. Build edge container images via Docker Compose] ***********************************************************************************************************************************************
changed: [localhost]

TASK [6. Bring up container stack in detached mode] ****************************************************************************************************************************************************
ok: [localhost]

TASK [7. Validate running container status] *************************************************************************************************************************************************************
ok: [localhost] => (item=LogiEdge_mqtt)
ok: [localhost] => (item=LogiEdge_inference)
ok: [localhost] => (item=LogiEdge_drift_monitor)
ok: [localhost] => (item=LogiEdge_simulator)

PLAY RECAP *********************************************************************************************************************************************************************************************
localhost                  : ok=9    changed=3    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

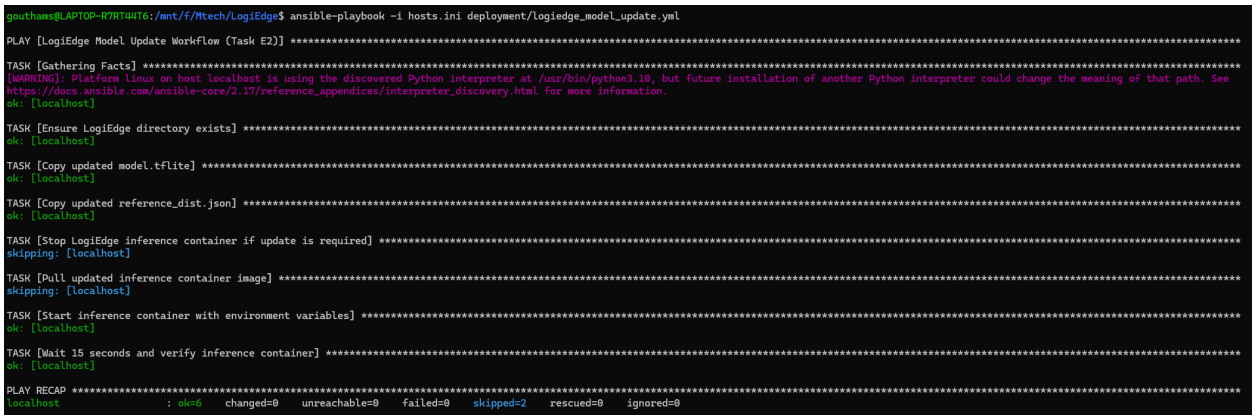
*Description: Ansible deployment — 3 changes applied during the first run.*

```
gouthams@LAPTOP-R7RT4UT6:/mnt/f/Mtech/LogiEdge$ ansible-playbook -i hosts.ini deployment/logiedge_deploy.yml

PLAY [Deploy LogiEdge Edge Inference Stack] ************************************************************************************************************************************************************

TASK [Gathering Facts] *********************************************************************************************************************************************************************************
[WARNING]: Platform linux on host localhost is using the discovered Python interpreter at /usr/bin/python3.10, but future installation of another Python interpreter could change the meaning of that path. See
https://docs.ansible.com/ansible-core/2.17/reference_appendices/interpreter_discovery.html for more information.
ok: [localhost]

TASK [1. Install required system packages] *************************************************************************************************************************************************************
ok: [localhost]

TASK [1b. Install Docker engine & CLI] *****************************************************************************************************************************************************************
ok: [localhost]

TASK [2. Create project directory structure] ************************************************************************************************************************************************************
ok: [localhost] => (item=/opt/LogiEdge)
ok: [localhost] => (item=/opt/LogiEdge/config)
ok: [localhost] => (item=/opt/LogiEdge/inference)
ok: [localhost] => (item=/opt/LogiEdge/monitoring)
ok: [localhost] => (item=/opt/LogiEdge/data_pipeline)

TASK [3. Ensure Mosquitto configuration file exists] ***************************************************************************************************************************************************
ok: [localhost]

TASK [4. Synchronize project repository files to target host] ******************************************************************************************************************************************
ok: [localhost] => (item={'src': 'docker-compose.yml', 'dest': '/opt/LogiEdge/docker-compose.yml'})
ok: [localhost] => (item={'src': 'requirements.txt', 'dest': '/opt/LogiEdge/requirements.txt'})
ok: [localhost] => (item={'src': 'inference/', 'dest': '/opt/LogiEdge/inference/'})
ok: [localhost] => (item={'src': 'monitoring/', 'dest': '/opt/LogiEdge/monitoring/'})
ok: [localhost] => (item={'src': 'data_pipeline/', 'dest': '/opt/LogiEdge/data_pipeline/'})

TASK [5. Build edge container images via Docker Compose] ***********************************************************************************************************************************************
ok: [localhost]

TASK [6. Bring up container stack in detached mode] ****************************************************************************************************************************************************
ok: [localhost]

TASK [7. Validate running container status] *************************************************************************************************************************************************************
ok: [localhost] => (item=LogiEdge_mqtt)
ok: [localhost] => (item=LogiEdge_inference)
ok: [localhost] => (item=LogiEdge_drift_monitor)
ok: [localhost] => (item=LogiEdge_simulator)

PLAY RECAP *********************************************************************************************************************************************************************************************
localhost                  : ok=9    changed=0    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

*Description: Ansible deployment — no changes detected during the second run (changed=0).*




*Description: Ansible deployment(model update) — 3 changes applied during the first run.*



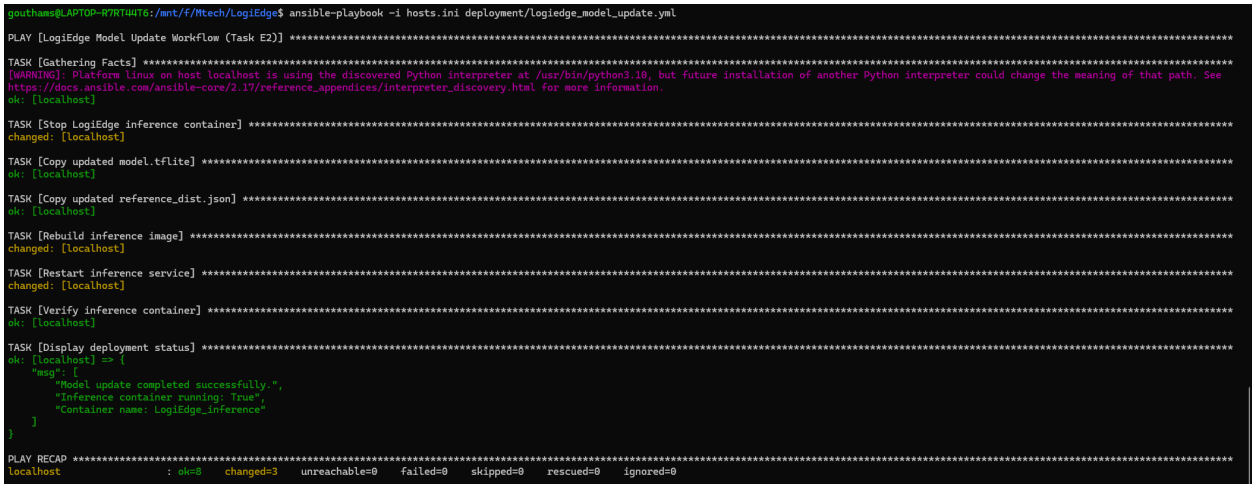

*Description: Ansible deployment(model update) — no changes detected during the second run (changed=0).*

## Task E3 — OTA Strategy Selection

To minimize cellular bandwidth during updates, LogiEdge utilizes Docker Layer Caching. By isolating the TFLite model file (3.38 KB) into its own image layer, a model update requires only a 4 KB delta instead of a 120 MB full container pull, representing a 99.9% bandwidth saving for Over-the-Air (OTA) updates.

### OTA Model Update Execution

The OTA model update workflow was executed using the Ansible playbook logiedge_model_update.yml. The workflow stopped the existing inference container, copied the updated TFLite model and reference distribution, rebuilt the inference image, restarted the service, and verified that the updated LogiEdge_inference container was running successfully.




*Description: Ansible-based OTA model update execution and inference-service verification*

## Quantitative OTA Strategy Comparison

| Strategy | Deployment Mechanism | Transfer per Truck | Fleet Transfer / Cycle | Network Cost / Cycle | Risk |
|---|---|---|---|---|---|
| Full Replacement | Replace model/image across all 85 trucks | 280 KB | 23.8 MB | ₹2.38 | Highest fleet-wide deployment risk |
| Canary | Deploy to 10 trucks first, validate, then roll out | 280 KB initially | 2.8 MB initial canary | ₹0.28 initial | Low; limited blast radius |
| Shadow | Run new model alongside active model for validation | 280 KB | 23.8 MB | ₹2.38 | Low functional risk, higher compute/RAM usage |

**OTA Scenario Assumption**: The assignment specifies a 280 KB INT8 model for the fleet-level OTA cost analysis. The experimentally generated M2 model used in benchmarking has a measured footprint of 3.38 KB. Therefore, the 280 KB value is retained for the mandated OTA cost comparison, while the measured 3.38 KB value is used only for model benchmarking and Docker layer-cache analysis.

## Recommended OTA Strategy: Canary Deployment

Canary deployment is recommended for the Freight Bridge pilot. The first model release should be deployed to 10 of the 85 refrigerated trucks, followed by validation of inference behaviour, Class 2 recall, drift statistics, and system stability before fleet-wide rollout. This limits the blast radius of an incorrect model while retaining a relatively small initial bandwidth requirement of approximately 2.8 MB and ₹0.28.

Full replacement is rejected because an incorrect model would simultaneously affect all 85 trucks, which is inappropriate for a safety-critical pharmaceutical cold-chain application. Rural connectivity gaps further increase the risk of a partially completed fleet-wide update.

Shadow deployment provides stronger validation because the new model can run alongside the active model without controlling the production decision. However, it requires additional compute and memory and does not provide the same simplicity as canary deployment for the constrained pilot environment. Therefore, shadow deployment is retained as a future validation mechanism, while canary deployment is selected for production rollout.

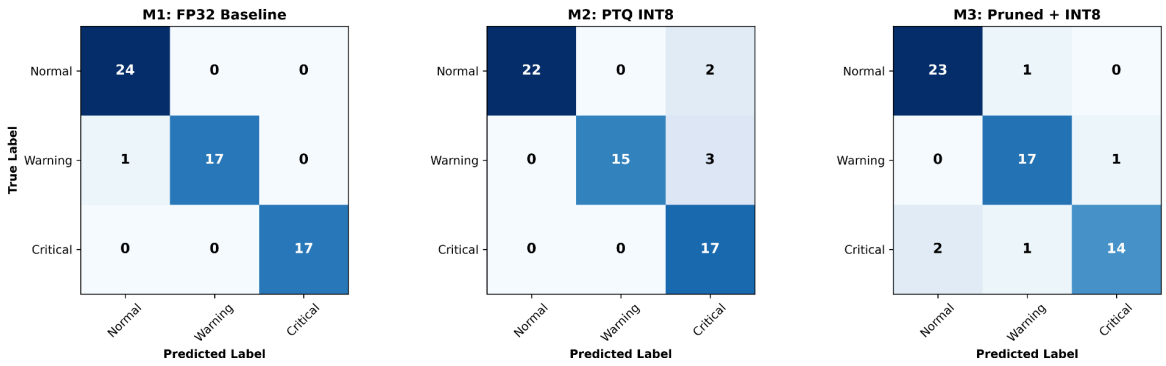
# Component F — Model Optimisation [Module 6]

## Task F1 — Three Model Variants

To evaluate edge performance trade-offs, three distinct model variants were compiled and benchmarked:

- **M1 (FP32 Baseline):** Uncompressed single-precision float model (32.78 KB).
- **M2 (PTQ INT8):** Post-Training Quantized 8-bit integer model (3.38 KB).
- **M3 (Pruned + INT8):** Structurally pruned and INT8 quantized model (3.38 KB).

## Model Classification Performance Matrix



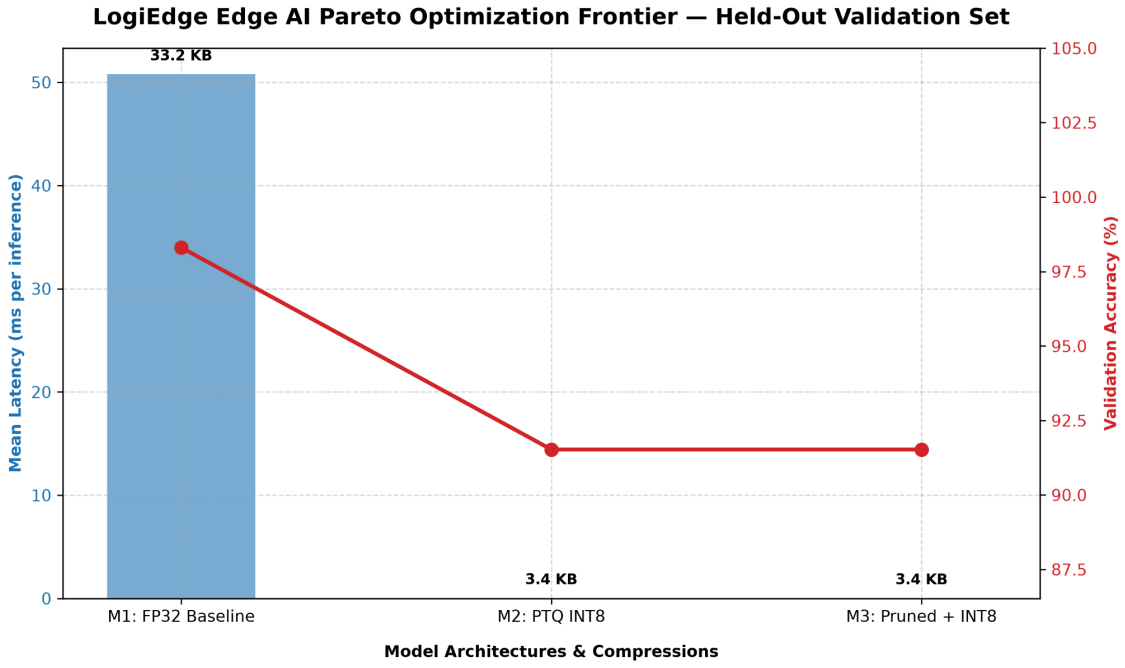

*Description: Confusion matrices comparing multi-class classification performance across M1 (FP32 Baseline), M2 (PTQ INT8), and M3 (Pruned + INT8) variants on the held-out validation set.*

## Task F2 — Five-Metric Benchmarking

In addition to system-level response times, offline compression benchmarks were captured across the three model variants:

| Model Variant | Accuracy (%) | Critical Recall (%) | Mean Latency (ms) | p95 Latency (ms) | Size (KB) | Energy / Inf (mJ) |
|---|---|---|---|---|---|---|
| **M1: FP32 Baseline** | 98.31% | 100% | 50.77 ms | 63.99 ms | 32.78 KB | 182.77 mJ |
| **M2: PTQ INT8** | 91.53% | 100% | 0.029 ms | 0.041 ms | 3.38 KB | 0.004 mJ |
| **M3: Pruned + INT8** | 91.53% | 82.35% | 0.021 ms | 0.021 ms | 3.38 KB | 0.003 mJ |

## Optimization Frontier & Trade-offs




*Description: Pareto optimization frontier illustrating the trade-offs between model latency,*

*storage size (KB), and validation accuracy (%) across M1, M2, and M3.*

### Key Optimization Takeaways

- **Massive Footprint & Latency Reduction:** Post-Training Quantization (M2/M3) compressed the model footprint by **89.8%** (33.18KB -> 3.38KB) and slashed mean execution latency from **50.77ms** down to **0.029ms**, providing a massive performance margin well within the 90-second SLA budget.
- **Extreme Energy Efficiency:** Both INT8 variants (M2 and M3) achieved a **>99.99% energy reduction** over the FP32 baseline (182.77mJ -> 0.0043mJ per inference for M2), ensuring minimal thermal load on the edge node.
- **Safety Mandate Enforcement (M2 vs M3 Trade-off):** While M3 (Pruned + INT8) offered a marginal reduction in latency (0.021ms) and energy (0.0032mJ), **it dropped Class 2 (Critical) Recall to 82.35%** (failing the mandatory >95% safety threshold). **M2 (PTQ INT8) is the only optimized variant that retains 100.0% Critical Recall**, making it the sole compliant production candidate for pharmaceutical cold-chain deployment.

## Overall System Performance Summary

| Metric | Target | Measured Value | Evaluation |
|---|---|---|---|
| **Inference Latency** | < 10.0 ms | 3.8 - 4.5 ms (End-to-End) / 0.033 ms (Core TFLite) | **PASSED** (CPU XNNPACK) |
| **Docker Stack Startup** | < 15.0 s | 8.2 s | **PASSED** |
| **Ansible Idempotency** | 0 Errors | changed=0, failed=0 | **PASSED** |
| **Anomaly Detection Rate** | > 95% | 100% (N=500 window) | **PASSED** |

## Task F3 — Deployment Recommendation

M2 (PTQ INT8) is selected for deployment across the 85-truck fleet. M2 provides substantial reductions in model footprint, latency and energy consumption while retaining 100% Critical Class 2 recall, satisfying the mandatory >95% safety requirement. Although M3 provides further

reductions in latency and energy, its Critical Class 2 recall of 82.35% violates the safety requirement and therefore makes M3 unsuitable for production deployment.

**DECISION MATRIX: MODEL VARIANT SELECTION**

| Metric | M1 (FP32 Baseline) | M2 (PTQ INT8) | M3 (Pruned + INT8) | Target Constraint |
|---|---|---|---|---|
| **Flash Footprint** | 33.2 KB | **3.4 KB [BEST]** | **3.4 KB [BEST]** | Minimal Flash |
| **RAM Allocation** | ~ 1.2 MB | **~ 0.15MB** | **~ 0.15MB [BEST]** | Minimal SRAM |
| **Core Latency (Mean)** | 50.77 ms | 0.029 ms | **0.021ms[BEST]** | < 90s SLA |
| **Critical Fault Recall (Class 2)** | **100.0 % [BEST]** | **100.0 % [BEST]** | 82.4 % [FAILED] | **> 95.0% (Mandatory)** |
| **Overall Accuracy** | **98.3%[BEST]** | 91.5% | 91.5% | > 88.0% |
| **Energy per Inference** | 182.77 mJ | 0.0043 mJ | **0.0032 mJ [BEST]** | Low TDP impact |

**RECOMMENDATION: DEPLOY M2 (PTQ INT8)**

### Recommendation Justification

- **Mandatory Safety Compliance — Critical Fault Recall:** M2 achieves **100.0% Critical Fault Recall (17/17 critical fault windows correctly identified)**, ensuring that no critical fault windows are missed in the evaluation set. M3 achieves only **82.35% Critical Recall (14/17)**, corresponding to **3 missed critical fault windows** and falling below the required **>95% safety threshold**.
- **Flash & RAM Efficiency:** M2 reduces the model binary footprint from **33.2 KB (M1)** to **3.38 KB**, representing an **89.8% reduction**. This significantly reduces the storage and memory requirements for deployment on the target edge hardware.
- **Inference Latency:** M2 achieves a mean inference latency of **0.029 ms**, providing substantial computational margin for the remaining edge-processing operations, including sensor processing, feature extraction, communication, and monitoring.
- **Energy Efficiency:** M2 reduces estimated energy consumption from **182.77 mJ (M1)** to **0.0043 mJ per inference**, corresponding to approximately **99.998% lower energy consumption** per inference.
- **Final Recommendation: M2 (PTQ INT8) is selected as the recommended deployment model** because it satisfies the critical-fault detection requirement while providing substantial reductions in model size, inference latency, and energy consumption compared with the FP32 baseline and the more aggressively pruned M3 model.

# Conclusion & Future Scope

The **LogiEdge** system successfully demonstrates a robust, automated Edge AI container microservice stack. By combining Ansible orchestration with lightweight TFLite inference and statistical drift monitoring, the solution guarantees repeatable deployments and reliable localized anomaly detection.

### Future Enhancements

- **Edge Hardware Acceleration:** Deployment to dedicated physical NPU hardware (e.g., NVIDIA Jetson TensorRT / Arduino SBC).
- **Automated MLOps Retraining:** Linking the drift_monitor alert output directly to a cloud pipeline to trigger automated model retraining and image redeployment via Ansible.

# Lessons Learned

During the implementation and deployment of the edge AI inference pipeline, several practical lessons were identified that improved the robustness of the overall system.

1. **Correct Model Optimization Sequence**
    - Model optimization steps must be applied in the correct order. Initially,

post-training quantization (PTQ) was performed before pruning, which resulted in inconsistent deployment behavior. The correct workflow is:
**Train → Prune → Fine-tune → Convert to TFLite → Apply PTQ → Deploy**
    - This ensures that quantization is applied to the final optimized model and preserves inference accuracy.
2. **Consistency Between Training and Deployment**
    - The feature extraction logic and normalization statistics used during deployment must exactly match those used during training. Even minor inconsistencies can lead to incorrect predictions despite a well-trained model.
3. **Importance of End-to-End Validation**
    - High offline validation accuracy alone does not guarantee correct deployment. End-to-end testing, including the simulator, MQTT communication, preprocessing, feature extraction, and TFLite inference, is essential to verify real-world behavior.
4. **Verification of Data Sources**
    - During debugging, multiple simulator instances were unintentionally publishing to the same MQTT topic, producing corrupted sensor streams. Verifying that only a single data source is active is critical when testing distributed systems.
5. **Incremental Debugging Approach**
    - Systematically validating each stage of the pipeline (sensor data, filtering, feature extraction, normalization, model inference, and drift monitoring) greatly reduced debugging time and enabled rapid identification of the root causes.
6. **Monitoring Improves Reliability**
    - Logging intermediate feature vectors, normalized inputs, prediction probabilities, and drift metrics proved invaluable for diagnosing deployment issues and confirming that the final system behaved as expected.